电路理论 mooc 答案——第十六章

解析

1.解:

$$F_{rms}=\left[\frac{1}{2\pi}\int_0^{\pi}(F_m\sin t)^2dt\right]^{\frac{1}{2}}=F_m\left(\int_0^{\pi}\frac{1-\cos 2t}{4\pi}dt\right)^{\frac{1}{2}}=\frac{F_m}{2}$$

$$\overline{F}=\frac{1}{2\pi}\int_0^{\pi}F_m\sin t dt=\frac{F_m}{2\pi}(-\cos t)|_0^{\pi}=\frac{F_m}{\pi}$$

选C

2.解:

$(a)\ u_S=u_{S0}+u_{S1}=5+5\sin t$

$$u_{rms}=\left[\frac{1}{2\pi}\int_0^{2\pi}\left(\frac{5+5\sin t}{10}\right)^2dt\right]^{\frac{1}{2}}=\frac{\sqrt{6}}{4}$$

$(b)\ u_S=u_{S0}+u_{S1}=-5+5\sin t\le 0\Rightarrow u_{rms}=0$

选D

3.解:

$If\ \omega=100rad/s,\ \dot{U}=10\angle 0°V,\ then\ \dot{I}_L=\dfrac{\dot{U}}{j\omega L}=0.1\angle-90°A,\ \dot{I}_C=j\omega C\dot{U}=0.1\angle 90°A.$

$If\ \omega=500rad/s,\ \dot{U}=3\angle 0°V,\ then\ \dot{I}_L=\dfrac{\dot{U}}{j\omega L}=0.006\angle-90°A,\ \dot{I}_C=j\omega C\dot{U}=0.15\angle 90°A.$

$Therefore,\ i_L=0.1\sin(100t-90°)+0.006\sin(500t-90°)A,\ \ i_C=0.1\sin(100t+90°)+0.15\sin(500t+90°)A,$

选A

4.解:

显然，受控源等效为$2R$的电阻。接下来利用叠加定理:

$(1)\ u_{S1}=1.5V,\ then\ u_R=\dfrac{R}{R+2R}u_{S1}=0.5V$

$(2)\ \dot{U}_{S2}=5\sqrt{2}\angle 90°V,\ then\ \dot{U}_R=\dfrac{R}{R+2R+j\omega L}\dot{U}_{S2}=\sqrt{2}\angle 36.87°V$

电压源的发出功率等于$2R+R$吸收的功率，$P=U_0I_0+\sum\limits_k U_kI_k\cos\varphi=(3*0.5)*0.5+(3*1)*1=3.75W$

$(3)\ \dot{I}_{S2}=2\angle 0°A,\ then\ \ \dot{U}_R=\dfrac{j\omega L}{3R+j\omega L}\dot{I}_{S2}\cdot R=\sqrt{2}\angle 45°V$

$Therefore,\ u_R=0.5+\sqrt{2}\sin(2t+36.87°)+\sqrt{2}\sin(1.5t+45°)V$

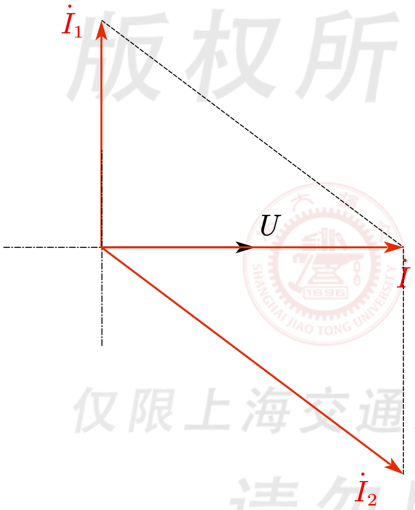
选B

5.解:

(1) L_1C_1 并联谐振，三次谐波等效开路,无法传到 R;

L_2C_2 串联谐振，基波等效短路接地,无法传到 R，但 C_2 上会有端电压。

(2) 如图,



选 C

$Remark$:谐振即电路呈电阻性。

6.解:

T 型去耦. $Let\ \dot{U}_S = 12\angle 0° V,\ then\ \dot{I} = \dfrac{\dot{U}_S}{j\omega[(4-2)+(4-2)//2]} \cdot \dfrac{1}{2} = -j\ A$

选 A

7.解:

(1) 受控电流源等效 $(3*25)\mu F$ 电容。

$$Let\ \dot{U}_S = 1\angle 0° V,\ Z_{eq} = 1 + Z//\left(j\omega L + \frac{1}{j\omega 4C}\right) = 1 + Z//0 = 1\Omega$$

$$then\ \dot{I} = 1\angle 0° A,\ P = \mathrm{Re}\left\{\dot{U}_S \dot{I}^*\right\} = 1W$$

(2) $I_{A1} = 0$，说明 $10\mu F, 0.1H$ 发生并联谐振，$\omega = \dfrac{1}{\sqrt{L_2C_2}} = 1krad/s$.

$$Let\ \dot{U}_S = 440\angle 0° V,\ \dot{I}_{A2} = \dot{U}_S \cdot \frac{\frac{1}{j\omega C_1} + R}{R + j\omega L_1 + \frac{1}{j\omega C_1} + R} \cdot \frac{1}{j\omega L_2} = -2\sqrt{2}\angle 45° A.$$

选 A

8.解:

(1) 标准的二表法测功率，注意桥接VW两相的电压端子接反的。

(2) $Let\ \dot{U}_U = \dfrac{U_L}{\sqrt{3}}\angle 0°,\ \dot{I}_U = I_L\angle(-\varphi),\ then$

$P = \mathrm{Re}\left\{\dot{U}_{VW}\dot{I}_U{}^*\right\} = \mathrm{Re}\{U_L\angle 90°\, I_L\angle\varphi\} = U_L I_L \sin\varphi = 120W.$

选A